that the lifetimes of such brightening, predicted from the physics of the interstellar gas, *are* a few minutes—and the chance of reacquiring the signal is small. We should really be pointing steadily at these coordinates in the sky, watching them for months.

Despite the fact that none of these signals repeats, there's an additional fact about them that, every time I think about it, sends a chill down my spine: 8 of the 11 best candidate signals lie in or near the plane of the Milky Way Galaxy. The five strongest are in the constellations Cassiopeia, Monoceros, Hydra, and two in Sagittarius —in the approximate direction of the center of the Galaxy. The Milky Way is a flat, wheel-like collection of gas and dust and stars. Its flatness is why we see it as a band of diffuse light across the night sky. That's where almost all the stars in our galaxy are. If our candidate signals really were radio interference from Earth or some undetected glitch in the detection electronics, we shouldn't see them preferentially when we're pointing at the Milky Way.

But maybe we had an especially unlucky and misleading run of statistics. The probability that this correlation with the galactic plane is due merely to chance is less than half a percent. Imagine a wall-size map of the sky, ranging from the North Star at the top to the fainter stars toward which the Earth's south pole points at the bottom. Snaking across this wall map are the irregular boundaries of the Milky Way. Now suppose that you were blindfolded and asked to throw five darts at random at the map (with much of the southern sky, inaccessible from Massachusetts, declared off limits). You'd have to throw the set of five darts more than 200 times before, by accident, you got them to fall as closely within the precincts of the Milky Way as the five strongest META signals did. Without repeatable signals, though, there's no way we can conclude that we've actually found extraterrestrial intelligence.

Or maybe the events we've found are caused by some new kind of astrophysical phenomenon, something that nobody has thought of yet, by which not civilizations, but stars or gas clouds (or something) that do lie in the plane of the Milky Way emit strong signals in bafflingly narrow frequency bands.